

Task 1

```
student@cse:~/OS_Lab_Assign_1_2401CS18$ chmod +x system_snapshot.sh
student@cse:~/OS_Lab_Assign_1_2401CS18$ ./system_snapshot.sh
===== SYSTEM SNAPSHOT =====
User           : student
Host name      : cse
Working directory : /home/student/OS_Lab_Assign_1_2401CS18
Login shell    : /bin/bash
Date and time  : Wednesday 29 July 2026 03:55:35 PM IST
Kernel release : 6.8.0-124-generic
Logged-in sessions : 1

Disk usage:


| Filesystem | Size | Used | Avail | Use% | Mounted on |
|------------|------|------|-------|------|------------|
| /dev/sda6  | 180G | 56G  | 115G  | 33%  | /          |



Non-hidden directory entries:
system_snapshot.sh : File
```

Task 2

```
student@cse:~/OS_Lab_Assign_1_2401CS18$ chmod +x number_inspector.sh
student@cse:~/OS_Lab_Assign_1_2401CS18$ ./number_inspector.sh 565
Original : 565
Reverse  : 565
Digits   : 3
Digit sum: 16
Palindrome: yes
student@cse:~/OS_Lab_Assign_1_2401CS18$
```

Task 3

```
student@cse:~/OS_Lab_Assign_1_2401CS18$ ./range_report.sh 3 -2
Range: 3 2 1 0 -1 -2
Count: 6
Sum: 3
Even: 3
Odd: 3
student@cse:~/OS_Lab_Assign_1_2401CS18$
```

Task 4 (Error Testing)

```
student@cse:~/OS_Lab_Assign_1_2401CS18$ chmod +x mini_os_toolkit.sh
student@cse:~/OS_Lab_Assign_1_2401CS18$ ./mini_os_toolkit.sh process 99999999
Error: Process with PID 99999999 does not exist or cannot be inspected.
student@cse:~/OS_Lab_Assign_1_2401CS18$ echo "Exit status: $?"
Exit status: 2
student@cse:~/OS_Lab_Assign_1_2401CS18$ ./mini_os_toolkit.sh process abc
Error: PID must be a positive integer.
student@cse:~/OS_Lab_Assign_1_2401CS18$ echo "Exit status: $?"
Exit status: 1
student@cse:~/OS_Lab_Assign_1_2401CS18$ printf 'First line\nSecond line\nThird line\n' > toolkit_test.txt
./mini_os_toolkit.sh file toolkit_test.txt
echo "Exit status: $?"
File name      : toolkit_test.txt
Size           : 34 bytes
Owner          : student
Permissions    : -rw-rw-r--
Line count     : 3
Exit status: 0
student@cse:~/OS_Lab_Assign_1_2401CS18$ ./mini_os_toolkit.sh file missing.txt
echo "Exit status: $?"
Error: File 'missing.txt' does not exist or is not a readable regular file.
Exit status: 2
student@cse:~/OS_Lab_Assign_1_2401CS18$ ./mini_os_toolkit.sh unknown
echo "Exit status: $?"
Error: Unknown command 'unknown'.
Usage: ./mini_os_toolkit.sh COMMAND [ARGUMENT]
Available commands:
  system      Display system information
  process PID Display information about a process
  file PATH   Display information about a file
  help        Display this help message
Exit status: 1
student@cse:~/OS_Lab_Assign_1_2401CS18$ ^[[200~./mini_os_toolkit.sh system extra
bash: ./mini_os_toolkit.sh: No such file or directory
student@cse:~/OS_Lab_Assign_1_2401CS18$ ./mini_os_toolkit.sh system extra
echo "Exit status: $?"
Error: The system command does not accept an argument.
Usage: ./mini_os_toolkit.sh COMMAND [ARGUMENT]
Available commands:
  system      Display system information
  process PID Display information about a process
  file PATH   Display information about a file
  help        Display this help message
Exit status: 1
```

Task 4 (Function execution)

```
student@cse:~/OS_Lab_Assign_1_2401CS18$ ./mini_os_toolkit.sh system; echo "Exit status: $?"
```

```
User       : student
Host name   : cse
Working dir : /home/student/OS_Lab_Assign_1_2401CS18
Login shell : /bin/bash
Kernel release : 6.8.0-124-generic
Date and time : Wednesday 29 July 2026 04:17:39 PM IST
Exit status: 0
```

```
student@cse:~/OS_Lab_Assign_1_2401CS18$ ./mini_os_toolkit.sh process $$; echo "Exit status: $?"
```

```
PID        : 3096
Parent PID: 3078
Owner      : student
State      : Ss
Command    : bash
Exit status: 0
```

```
student@cse:~/OS_Lab_Assign_1_2401CS18$ ./mini_os_toolkit.sh file mini_os_toolkit.sh; echo "Exit status: $?"
```

```
File name   : mini_os_toolkit.sh
Size        : 2869 bytes
Owner       : student
Permissions : -rwxrwxr-x
Line count  : 109
Exit status: 0
```

```
student@cse:~/OS_Lab_Assign_1_2401CS18$ ./mini_os_toolkit.sh help; echo "Exit status: $?"
```

```
Usage: ./mini_os_toolkit.sh COMMAND [ARGUMENT]
```

```
Available commands:
```

system	Display system information
process PID	Display information about a process
file PATH	Display information about a file
help	Display this help message

```
Exit status: 0
```

```
student@cse:~/OS_Lab_Assign_1_2401CS18$
```